

Grad Café SQL Query Analysis

Module 3 — Query Documentation

1. How many entries do you have in your database who have applied for Fall 2026?
 - a. Counts every row in the applicants table where the term column equals 'Fall 2026'. It uses COUNT(*) with a WHERE filter to extract entries for just Fall 2026
 - b. COUNT(*) is used because it extracts all rows regardless of NULL. An exact string match to 'Fall 2026' is appropriate
2. What percentage of entries are from international students (not American or Other) (to two decimal places)?
 - a. Calculates the percentage of international applicants out of all applicants. It uses COUNT(*) FILTER to count only matching rows, divides by the total count, and does appropriate rounding to get percentage
 - b. We are counting all applicants, so COUNT(*) is used, and FILTER (WHERE citizenship = 'International') is appropriate since it does string matching
3. What is the average GPA, GRE, GRE V, GRE AW of applicants who provide these metrics?
 - a. Calculates the average of all four score columns, but only over rows where at least one score was reported. The AVG() function automatically ignores NULL values, so applicants who did not report a given score are excluded from that specific average calculation.
 - b. The '::numeric' is needed because AVG() returns a double precision type and ROUND() needs a numeric type to round to a specific number of decimals. The WHERE <metric> IS NOT NULL guarantees averages are calculated from applicants that provided those metrics
4. What is their average GPA of American students in Fall 2026?
 - a. Filters American students applying for Fall 2026 who reported a GPA, then takes the average (accomplished using WHERE citizenship == 'American'). All three conditions must be true simultaneously, so we employ AND to combine them.
5. What percentage of entries for Fall 2026 are Acceptances (to two decimal places)?
 - a. Using WHERE, we filter for Fall 2026 entries using string matching of the semester. Using COUNT(*), extract the number of all entries, then calculate the percentage of those entries with status 'Accepted', using WHERE 'status' == 'Accepted'.
6. What is the average GPA of applicants who applied for Fall 2026 who are Acceptances?
 - a. Three conditions must be true at the same time: semester is Fall 2026, the status is 'Accepted', and the applicant provided a GPA. This can be accomplished using

the WHERE function. The average GPA is calculated using AVG of these filtered applicants

- b. The ::numeric function is needed to rounded to the appropriate number of decimal places
7. How many entries are from applicants who applied to JHU for a master's degrees in computer science?
 - a. The university and program of each entry is combined into one string, so ILIKE with 'wildcards' to search within each string. The degreeType is checked using the same method to account for differences between degreeType entries 'Master' and 'Masters'. Both of these options are valid
 - b. We then COUNT(*) from applicants that meet this criteria.
8. How many entries from 2026 are acceptances from applicants who applied to Georgetown University, MIT, Stanford University, or Carnegie Mellon University for a PhD in Computer Science?
 - a. To get this information, we get the COUNT(*) from applicants that meet our criteria. We use ILIKE ANY (ARRAY[...]) to look at the list of universities contained within the university-program string, and select any one of them. Next, since multiple criteria must be true at once, we use the AND operator and ILIKE to filter for the Computer Science program, PhD degree type, and the semester in the year 2026. Status of accepted is found using a string match.
9. Do your numbers for question 8 change if you use LLM Generated Fields (rather than your downloaded fields)?
 - a. This query repeats the same logic as question 8 but uses the llm_generated_university and llm_generated_program columns instead of the combined university-program field.
 - b. The LLM- generated fields may capture more information that the raw entries because some of the user-input entries may use abbreviations or have misspellings that will not be caught while using the university-program field
10. What is the PhD rejection rate in 2025 vs 2026?
 - a. Gets PhD applicants by term across 2025 and 2026 Fall and Spring semesters, then calculates the rejection rate within each group.
 - b. GROUP BY term splits results into one row per semester so we can compare rejection rates across different semesters. Then we FILTER specifically for PhD applicants who got rejected, count this number, and then divide this number by the total number of applicants. This metric gives information whether PhD programs became more selective/competitive between 2025 and 2026.
11. What is the average GPA of accepted vs rejected PhD applicants in 2026?
 - a. Calculates the average GPA of PhD applicants who were accepted against those who were rejected in 2026. Only applicants who reported a GPA are included.

- b. Grouping provides the ability to compute a separate average GPA for rejected and accepted applicants, excluding Waitlisted and Interview entries. The gpa IS NOT NULL filter ensures we only average over applicants who self-reported their GPA.